

Question ID: d84a514a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Easy

Question

The function $f(x) = 240,000(1.22)^x$ gives a company’s predicted annual revenue, in dollars, x years after the company started selling jewelry online, where $0 < x \leq 10$. What is the best interpretation of the statement “ $f(5)$ is approximately equal to 648,650” in this context?

Answer

- A. 5 years after the company started selling jewelry online, its predicted annual revenue is approximately 648,650 dollars.
- B. 5 years after the company started selling jewelry online, its predicted annual revenue will have increased by a total of approximately 648,650 dollars.
- C. When the company’s predicted annual revenue is approximately 648,650 dollars, it is 5 times the predicted annual revenue for the previous year.
- D. When the company’s predicted annual revenue is approximately 648,650 dollars, it is 5% greater than the predicted annual revenue for the previous year.

Correct Answer: A

Rationale

Choice A is correct. It’s given that the function f gives a company’s predicted annual revenue, in dollars, x years after the company started selling jewelry online. Since the value of $f(5)$ is the value of $f(x)$ when $x = 5$, it follows that “ $f(5)$ is approximately equal to 648,650” means that $f(x)$ is approximately equal to 648,650 when $x = 5$. Therefore, the best interpretation of the given statement is that 5 years after the company started selling jewelry online, its predicted annual revenue is approximately 648,650 dollars.

Choice B is incorrect. The function f gives the predicted annual revenue, not the predicted increase in annual revenue.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect. In the given function, x represents the number of years after the company started selling jewelry online, not the percent increase in revenue from the previous year.